Exercise: 2.3

1. $\frac{a^x}{\log a}+a^a x-\frac{x^{n+1}}{n+1}+c$
2. $\frac{1}{a^x \log a}-\frac{1}{b^x \log b}+c$
3. $e^x+e^{2x}+\frac{e^{3x}}{3}+c$
4. $\frac{e^{2x}}{2}+\frac{e^{-4x}}{4}+c$
5. $\frac{e^{4x}}{4}+c$
6. $e^{\left(x+\frac{1}{x}\right)}+c$
7. $-\frac{1}{\log x}+c$
8. $c=1, f(x)=e^x+1$

Exercise: 2.4

1. $2\sin x+3\cos x+4\tan x+5\cot x+c$
2. $-\frac{3}{4}\cos x+\frac{1}{12}\cos 3x+c$
3. $\tan x+c$
4. $\tan x-\cot x+c$
5. $-[\sin x+\cos x]+c$

Exercise: 2.5

1. $-e^{-x}(x+1)+c$
2. $e^{3x}\left[\frac{x^3}{3}-\frac{x^2}{3}+\frac{2x}{9}-\frac{2}{27}\right]+c$
3. $x(\log x-1)+c$
4. $\frac{x^2}{2}\left[\log x-\frac{1}{2}\right]+c$
5. $\frac{x^{n+1}}{n+1}\left(\log x-\frac{1}{n+1}\right)+c$
6. $\frac{e^{x^2}}{2}\left(x^4-2x^2+2\right)+c$

Exercise: 2.6

1. $\log\left|x^2+5x-7\right|+c$
2. $\frac{1}{4}\log\left|x^4+1\right|+c$
3. $\frac{1}{2}\log\left|e^{2x}-2\right|+c$
4. $\frac{(\log x)^4}{4}+c$
5. $2\sqrt{3x^2+7x-1}+c$
6. $\frac{4}{3}\left(x^2+x+1\right)^{\frac{3}{2}}+c$
7. $\frac{1}{54}\left(1+x^9\right)^6+c$
8. $\frac{1}{e}\log\left|x^e+e^x\right|+c$
9. $\log\left|\log x\right|+c$
10. $\frac{1}{10}\log\left|\frac{x^2-2}{2x^2+1}\right|+c$
11. $xe^x\left[\log\left(xe^x\right)-1\right]+c$
12. $\log|x|-\frac{1}{2}\log\left|x^2+1\right|+c$
13. $\frac{e^x}{x^2}+c$
14. $\frac{e^x}{(x+1)^2}+c$
15. $\frac{e^{3x}}{9x}+c$

Exercise 2.7

1. $\frac{1}{24}\log\left|\frac{3+4x}{3-4x}\right|+c$
2. $\frac{1}{10}\log\left|\frac{9+x}{1-x}\right|+c$
3. $\frac{1}{6\sqrt{2}}\log\left|\frac{\sqrt{2}x-3}{\sqrt{2}x+3}\right|+c$
4. $\frac{1}{3}\log\left|\frac{x-2}{x+1}\right|+c$
5. $\log\left|\frac{x+1}{x+2}\right|+c$
6. $\frac{1}{10}\log\left|\frac{x-1}{x+4}\right|+c$
7. $\frac{1}{6}\log\left|\frac{e^x-3}{e^x+3}\right|+c$
8. $\frac{1}{3}\log\left|3x+\sqrt{9x^2-7}\right|+c$
9. $\log\left|(x+3)+\sqrt{x^2+6x+13}\right|+c$
10. $\log\left|\left(x-\frac{3}{2}\right)+\sqrt{x^2-3x+2}\right|+c$
11. $\frac{1}{4}\log\left|x^4+\sqrt{x^8-1}\right|+c$

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